

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Medium

Time allowed: 10

Score: /5

Percentage: /100

Question 1

Methyl iodide can be heated with ammonia. Which of the options describes the products of this reaction?

- A. Methylamine
- B. Dimethylamine
- C. Trimethylamine
- D. All of the above

[1 mark]**Question 2**

Which of the following reactions would produce 2-hydroxy-2-methylpropanenitrile?

- A. Ethanal + Sulphuric acid
- B. Propanone + Hydrogen cyanide
- C. Ethanal + Ammonia
- D. Propanone + Ammonia

[1 mark]**Question 3**

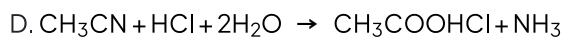
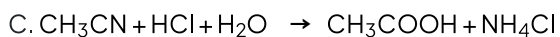
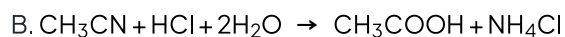
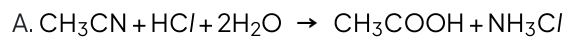
Reactions were carried out to produce nitriles. Which combination of reactants and reagents is not correct?

	Reactant	Reagent
A	Halogenoalkane	Potassium cyanide and ethanol
B	Amide	Phosphorus (V) oxide
C	Aldehyde	Hydrogen cyanide
D	Ketone	Ammonia

[1 mark]

Question 4

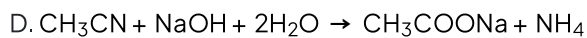
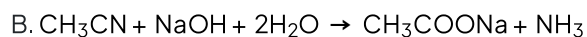
The following equations show reactions with nitrogen compounds. Which one correctly shows the hydrolysis of nitriles with dilute acid?



[1 mark]

Question 5

The following equations show reactions with nitrogen compounds. Which one correctly shows the hydrolysis of nitriles with dilute sodium hydroxide?



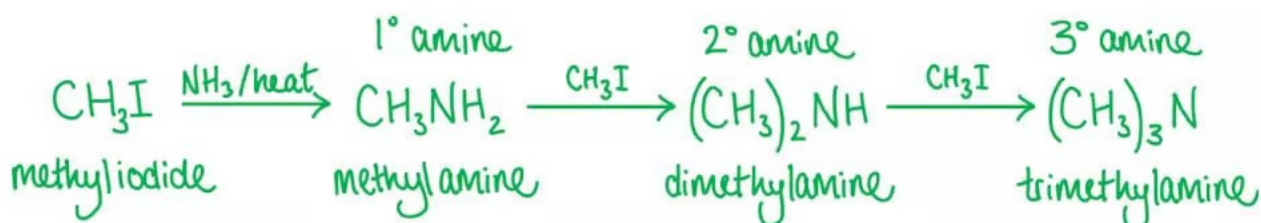
[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- The reaction of halogenoalkanes with **ammonia** produces primary amines.
- Ammonia is a **nucleophile** and will replace the iodine to form the primary amine **methylamine**.
- The nitrogen in methylamine can act as a **nucleophile** and react with another methyl iodide molecule to form a **secondary amine (dimethylamine)**.
- The nitrogen in dimethylamine can again act as a nucleophile and react with another methyl iodide molecule to form a **tertiary amine (trimethylamine)**.
- The multiple substitution reaction is:



To prevent a **mixture** of amines being formed, **excess** of **hot ethanolic ammonia under pressure** should be reacted with the halogenoalkane, this will result in the formation of primary amines **only**.

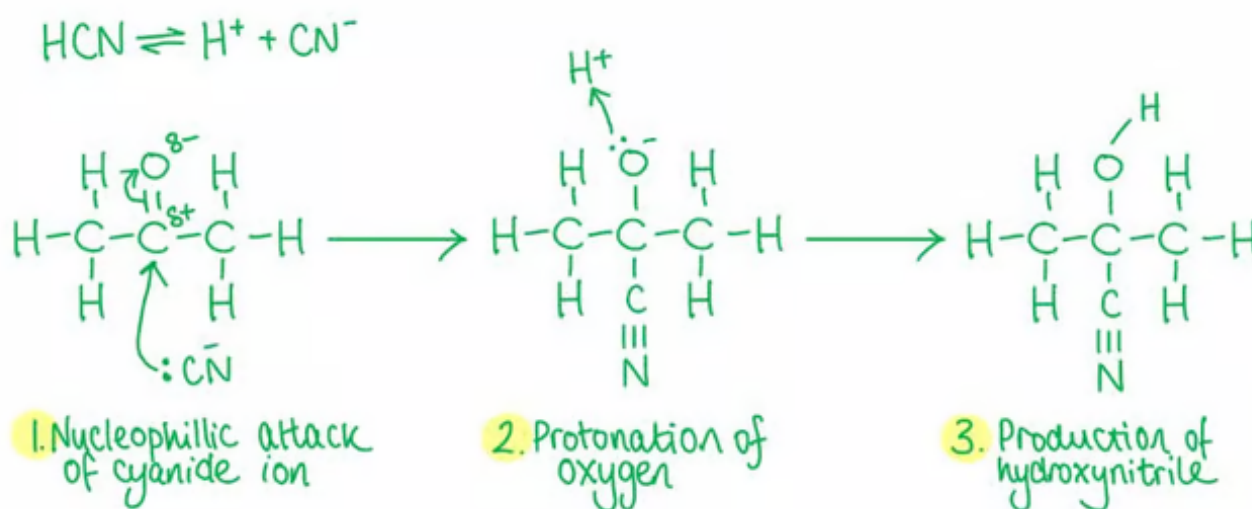
A, B & C are incorrect as a mixture of amines is produced as the question does not specify that the ammonia was in **excess**.

2

The correct answer is **B** because:

- 2-hydroxy-2-methylpropanenitrile is a **hydroxynitrile**.

- **Hydroxynitriles** can be formed from the **nucleophilic addition** reaction of **aldehydes** and **ketones** with **HCN**, **KCN** as catalyst and heat.
- A **nucleophilic addition** is when a **nucleophile** is added to an unsaturated bond to form a **single** product.
 - A **nucleophile** is a species that donates a pair of electrons.
- **Propanone** (which is a ketone) and hydrogen cyanide are the correct reagents required for the conversion of propanone to 2-hydroxy-2-methylpropanenitrile.
- The reaction mechanism is:



A, C & D are incorrect as though these reactions contain the correct reactant (either an **aldehyde** or **ketone**), they do not contain the correct reagent (**hydrogen cyanide**, **HCN**).

3

The correct answer is **D** because:

- To produce nitriles from aldehydes and ketones, **hydrogen cyanide** should be used.
- The reagent in option D should have been **hydrogen cyanide** and not ammonia.

A, B & C are incorrect as these are all correct combinations of reactants and reagents.

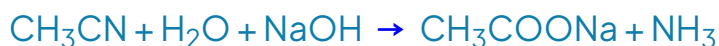
4

The correct answer is **B** because:

- **Hydrolysis** is the breakdown of a compound using water.
- **Nitriles** can be hydrolysed either by **refluxing** it with **dilute hydrochloric acid** or with **sodium hydroxide solution**.
- When hydrolysed with a dilute acid the -CN group at the end of the hydrocarbon chain is converted into a -COOH group, forming a **carboxylic acid**.
- Ammonia in solution will act as a **base** to form **ammonium salt** with Cl^- (NH_4Cl).
- Option **B** is therefore the only correct option.

When a nitrile is **refluxed** with **sodium hydroxide solution**, **hydrolysis** can also occur. Instead of forming an **ammonium salt**, now a **sodium salt** is formed, and **ammonia gas** is given off. The ammonia is formed from the reaction between **ammonium** (NH_4^+) ions and **hydroxide** (OH^-) ions

The reaction equation for this hydrolysis equation is:



To form the free **carboxylic acid** the final solution has to be acidified with a strong acid such as dilute **hydrochloric acid (HCl)** such that the **ethanoate** ion in the **sodium ethanoate salt** will react with the hydrogen ions to form **ethanoic acid**. Once reacted, ethanoic acid is a **weak acid** and holds onto the hydrogen rather than dissociating.

A is incorrect as **ammonium** chloride (NH_4Cl) is formed, not **ammonia** chloride (NH_3Cl).

C is incorrect as the water molecules are not correctly balanced in this reaction. 2 moles of water are needed for the reaction to become balanced.

D is incorrect as **ammonium** chloride (NH_4Cl) is formed, **not** ammonia (NH_3). Ammonia would have been formed if the nitrile was hydrolysed by a sodium hydroxide solution.

5

The correct answer is **A** because:

- **Hydrolysis** is the breakdown of a compound using water.

- **Nitriles** can be hydrolysed either by **refluxing** it with **dilute hydrochloric acid** or with **sodium hydroxide solution**.
- When hydrolysed with **sodium hydroxide solution**, instead of forming an **ammonium salt**, now a **sodium salt** is formed, and **ammonia gas** is given off.
- The ammonia is formed from the reaction between **ammonium** (NH_4^+) ions and **hydroxide** (OH^-) ions.
- Option **A** is therefore the correct option.

